Making NDFU bottom-up: the unified adoption rule and its two variants

NDFU internal design note

2026-08-11, updated 2026-11-05

Status: the unified rule is the engine of 05 (Rcpp since PR #4, §8).
Companion scripts: `02_network_calibrated.R`, `05_unified_diffusion_sweep.R`, `06*`, `07`, `09`

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1 Why this document

The Social Networks special-issue rejection (Bianchi's follow-up email) said:

> *"the underdevelopment of the behavioural model assumed in your ABM. While it can certainly be considered an agent-based model, the behaviour of the agents is relatively abstract, and the experimental design appears to be governed by macro-level manipulation rather than bottom-up dynamics."*

Two pieces of the current model are exposed to that critique:

1. **The network threshold τ_i is exogenous.** It is drawn from $\mathcal{N}(\tau_{\text{mean}}, \tau_{\text{sd}})$, truncated to $[0, 1]$, *independently of the agent's own measured preference* q_i. The sweep over $(\tau_{\text{mean}}, \tau_{\text{sd}})$ is exactly the "macro-level manipulation" Bianchi points at: a population-level dial with no micro-foundation in the survey data.
2. **The two channels are bolted together.** Rational adoption ($q_i \leq \Gamma$) and social adoption ($\tau_i \leq \tilde{E}_i$) are two separate rules joined by an OR. A referee can ask why an agent runs two unrelated decision procedures.

Tur, Zeppini & Frenken (2024, *Social Networks* 76:12–21) — published in our target journal — has a *single* micro rule (a utility comparison, where social reinforcement enters the utility calculus), but **no social-distance sensitivity in influence**: every adopting neighbor reinforces equally. NDFU's distinctive ingredient is exactly that sensitivity (Centola 2011). The question is therefore:

> Can we keep NDFU's selective (distance-sensitized) influence while adopting a Tur-style single utility-based decision rule — making the model bottom-up without giving up its identity?

Answer: yes. The instrument is a single **unified adoption rule** (§4), which admits two operationalizations — a *linear-fractional* variant and a *multiplicative γ-variant* that nests Tur (2024) exactly. Their pros and cons are compared in detail in §5.

2 Notation and the current NDFU model (baseline)

Symbol	Meaning	Source
$\Gamma \in [0, 1]$	Intrinsic Utility Level (IUL) of the behavior	swept
$q_i \in [0, 1]$	Minimum Utility Requirement (MUR): intrinsic aversion	**empirical** (GSS/ATP items)
$\tau_i \in [0, 1]$	network threshold (fraction of influential adopters needed)	**exogenous** $\mathcal{N}(\tau_{\text{mean}}, \tau_{\text{sd}})$
$d_{ij} \in [0, 1]$	Gower distance in Blau space (age, educ, race, relig, sex)	**empirical**
$h \in [0, 1]$	Maximum Social Proximity (MSP): tolerance to dissimilarity	swept (macro)
$a_j \in \{0, 1\}$	adoption state of j	dynamic
X_{ij}	adjacency	imputed ERGM networks

Current rules (as in the paper).

- Rational channel: adopt if $q_i \leq \Gamma$ (Eq. 1 of the paper).
- Social channel: adopt if $\tau_i \leq \tilde{E}_i$, with selective effective exposure

$$\tilde{E}_i \;=\; \frac{\sum_{j \neq i} X_{ij}\, \tilde{a}_j}{\sum_{j \neq i} X_{ij}}, \qquad \tilde{a}_j = 1 \;\iff\; a_j = 1 \wedge U_{ij} \leq \sigma(h - d_{ij}), \tag{1}$$

  where $U_{ij} \sim \text{Unif}(0, 1)$ is redrawn per step and $\sigma(z) = \left(1 + e^{-z/T}\right)^{-1}$ with $T = 0.02$ in the implementation (a near-hard gate at $d_{ij} = h$; see `scripts/04_GSS_diffusion_sims.R`, line 304).

Implementation nuance worth knowing (it matters for the comparison with Tur): in the code, the rational channel is implemented by setting the node's threshold to 10^{-6}, so a rational agent still needs **one influential contact** (exposure > 0) before adopting. The engine is therefore already a word-of-mouth model for *awareness*: low-q agents behave as simple contagion, not as spontaneous ($\varphi = 0$) adopters. The paper's Eq. 1 (adoption at exposure 0) is the idealization; the code is contact-gated.

3 Tur (2024) mechanism, for reference

- Word-of-mouth **percolation**: agents must be *informed* through the network before they can decide.
- Single decision rule: adopt when the innovation's intrinsic utility ξ meets the agent's minimum utility requirement.
- **Social reinforcement modifies the requirement** — exact published form (confirmed against both PDFs):

$$u_t^i \;=\; u_0^i \cdot \left(\frac{1}{a_t^i}\right)^{\gamma}, \tag{2}$$

  where u_0^i is the initial minimum utility requirement, $a_t^i \in \mathbb{N}$ the number of **adopting neighbors** of i at time t (a state count, not received recommendations; the rule is evaluated only once informed, so $a \geq 1$), and $\gamma \geq 0$ the social-reinforcement strength. The lowering is **multiplicative** (power-law): each doubling of adopting neighbors divides the requirement by the same factor 2^{γ}. With $\gamma = 0$ the model reduces to pure social percolation. The threshold-like behavior emerges from this lowering, not from a separate threshold parameter.
- Homophily ρ is purely **structural pre-wiring**: agents with similar u are placed together *before* diffusion. Influence itself is **distance-blind**: any adopting neighbor reinforces equally.
- Preferences are artificial distributions (Beta(α, β) family in 2018; uniform in 2024); networks are Watts–Strogatz with $N = 10{,}000$ and $k = 4$ neighbors on average.

NDFU vs. Tur, in one line: **Tur has the single bottom-up rule but uniform influence; NDFU has selective influence but a two-channel rule with an exogenous τ.** Each has what the other lacks.

4 The unified additive-utility rule

4.1 The rule

Replace both channels by one utility comparison per agent per step:

$$\text{adopt}_i(t) = 1 \iff \Gamma + \lambda\, \tilde{E}_i(t) \;\geq\; q_i. \tag{3}$$

- $\tilde{E}_i(t)$ is **exactly the current selective effective exposure** (1), unchanged. This is where social distance enters: only adopters passing the stochastic proximity gate $\sigma(h - d_{ij})$ count.
- $\lambda > 0$ is the **social-proof coupling**: how much situational utility a fully-adopted, fully-similar neighborhood adds, in the same units as Γ and q_i.

4.2 Two algebraically identical framings (choose the second)

1. **Requirement-lowering (Tur's framing):** $q_i^{\text{eff}}(t) = q_i - \lambda \tilde{E}_i(t)$; adopt when $\Gamma \geq q_i^{\text{eff}}(t)$. Preferences are *modified* during diffusion.

2. **Situational-utility (recommended framing):** $U_i(t) = \Gamma + \lambda \tilde{E}_i(t)$; adopt when $U_i(t) \geq q_i$. The preference q_i **never changes**; what changes is the utility the situation offers (peers add legitimacy, lower uncertainty, network externalities). This is literally Granovetter's threshold as a *situated cost–benefit crossing* (1978, p. 1422), and it preserves NDFU's declared contrast with Tur ("preferences are not modified during diffusion", `theoretical_reviewed.md`, entry 7).

4.3 The derived threshold — the two old channels fall out as limit cases

Rearranging (3), adoption requires $\tilde{E}_i(t) \geq \tau_i^*$ with the **derived threshold**

$$\tau_i^* \;=\; \max\left\{0,\; \frac{q_i - \Gamma}{\lambda}\right\}. \tag{4}$$

- $q_i \leq \Gamma \;\Rightarrow\; \tau_i^* = 0$: adopts at first influential contact — **the current rational channel** (Granovetter's low-threshold "instigator", exactly as currently implemented with $\tau = 10^{-6}$).
- $q_i > \Gamma \;\Rightarrow\; \tau_i^* > 0$: needs selective exposure proportional to the **utility deficit** — the social channel, now derived instead of postulated.
- $q_i > \Gamma + \lambda \;\Rightarrow\; \tau_i^* > 1$: **can never adopt** ($\tilde{E} \leq 1$) — a "structurally immune" agent. When exactly this occurs is worked out in §5.1.

4.4 What this buys and what it costs

Buys.

- **Bottom-up microfoundation.** One decision rule, all heterogeneity empirical: τ_i^* is a deterministic function of the *measured* q_i and the product (Γ, λ). The arbitrary $\mathcal{N}(\tau_{\text{mean}}, \tau_{\text{sd}})$ disappears. This is the direct answer to Bianchi.
- **Behavioral realism as a bonus:** resistance to peer influence is now positively correlated with intrinsic aversion (the same latent trait drives both) — defensible, and testable in principle.
- **Nesting.** Setting the gate to $w \equiv 1$ ($h \to \infty$, distance-blind) removes the distance selectivity and leaves a Tur-style (distance-blind) reinforcement model; setting $\lambda \to 0$ recovers a pure percolation-of-awareness model. Tur (2024)'s *exact* published form (2) is recovered by the γ-variant of the rule — see §4.6. NDFU's selectivity remains the *identified added ingredient*, cleanly positioned against a paper published in the target journal.
- **Implementability.** τ_i^* is static given (Γ, λ), so the existing `netdiffuseR` threshold machinery runs it as-is: `threshold.dist = pmax(0, (q - Γ)/λ)` replaces the Normal draw + the 10^{-6} hack. No engine changes needed.

Costs / changes.

- The rational/social decomposition becomes **ex-post bookkeeping** (adopters with $q_i \leq \Gamma$ vs. $q_i > \Gamma$) rather than two mechanisms — same tally we already compute, but the interpretation shifts.
- τ_{sd} disappears as a free dial; its large positive effect on cascades (heterogeneity creates susceptible minorities) is now carried by the *empirical* dispersion of q_i, scaled by $1/\lambda$. λ replaces $(\tau_{\text{mean}}, \tau_{\text{sd}})$ — the parameter space shrinks from 16 (4×4) threshold configurations to a 1-D λ sweep.
- The (IUL × MSP) heatmaps will not be identical: thresholds now co-vary with q. Whether the phase structure survives is an empirical question — answered: it does (see §7.1).

4.5 Where social distance enters (explicitly)

Only inside $\tilde{E}_i$, via the per-contact stochastic gate:

$$\Pr\big(j\text{'s adoption counts for } i\big) \;=\; \sigma\!\left(\frac{h - d_{ij}}{T}\right), \qquad T = 0.02. \tag{5}$$

Identical to the current model. The unified rule changes the *decision rule*, not the *influence filter*.

4.6 Recovering Tur (2024) exactly — the nesting, step by step

Tur's published rule (exact form, §3): an agent that has been *informed* through the network adopts when the product's utility meets its *current* requirement,

$$\xi \;\geq\; u_t^i, \qquad u_t^i = u_0^i \cdot \left(\frac{1}{a_t^i}\right)^{\gamma}, \tag{6}$$

with u_0^i the initial minimum utility requirement, $a_t^i \geq 1$ the number of adopting neighbors of i at time t (the rule is only evaluated once informed, so $a \geq 1$ and the expression is well-defined), and $\gamma \geq 0$ the social-reinforcement strength. $\gamma = 0$ gives $u_t = u_0$: pure social percolation (Solomon & Weisbuch 2000), Tur's own baseline.

Log-linearization — Tur IS an additive requirement-lowering rule, in logs. Taking logarithms (all quantities positive):

$$\ln \xi \;\geq\; \ln u_0^i \;-\; \gamma \,\ln a_t^i. \tag{7}$$

So the published reinforcement is additive with the concave gain $g(a) = \ln a$. This dictates the right unified rule: it must be additive in *log-utility*, i.e. **multiplicative in levels**.

The Tur-exact unified rule (γ-form). Define the distance-sensitized adopting *count*

$$\tilde{A}_i(t) \;=\; \sum_j X_{ij}\, \tilde{a}_j(t), \qquad \tilde{a}_j = 1 \;\iff\; a_j = 1 \,\wedge\, U_{ij} \leq \sigma\!\left(\frac{h - d_{ij}}{T}\right), \tag{8}$$

and let social proof *multiply* the situational utility:

$$\boxed{\text{adopt}_i(t) \;\iff\; \Gamma \cdot \big(\tilde{A}_i(t)\big)^{\gamma} \;\geq\; q_i} \qquad \text{(evaluated when } \tilde{A}_i \geq 1\text{)}. \tag{9}$$

Equivalently, in Tur's own requirement-lowering form: $q_{i,t} = q_i \cdot (1/\tilde{A}_i(t))^{\gamma}$, adopt when $\Gamma \geq q_{i,t}$. The preference q_i is fixed; what the similar adopters change is the *situational* requirement — the same framing as §4.2 (but see §5.5 for the substantive difference this form does carry).

Recovery, in one step. Switch off the proximity gate ($h \geq 1 + 5T \Rightarrow \sigma \to 1 \Rightarrow \tilde{a}_j = a_j$, so $\tilde{A}_i(t) = a_t^i$), and rule (9) reads

$$\Gamma \cdot (a_t^i)^{\gamma} \geq q_i \qquad \Longleftrightarrow \qquad \xi \;\geq\; u_0^i \cdot \left(\frac{1}{a_t^i}\right)^{\gamma} \tag{10}$$

under the identification

$$\boxed{\xi = \Gamma, \qquad u_0^i = q_i, \qquad \text{same } \gamma\,.} \tag{11}$$

The recovery is **exact on any graph**: Tur's reinforcement is a pure count, and the γ-form uses the count directly — no degree normalization is involved, so no degree-regularity caveat is needed (unlike the linear-fractional variant, which normalizes by degree and matches Tur's count reading only on degree-regular substrates).

Awareness / percolation layer (unchanged): the engine's contact-gating (the $\tau = 10^{-6}$ device, §2) means the rule is only evaluated at $\tilde{A}_i \geq 1$ — precisely Tur's "informed" condition with per-contact transmission probability 1; a probability $p < 1$ is emulated by a constant-acceptance gate ($\sigma \equiv p$).

Derived threshold (count form). Rearranging (9):

$$\text{adopt}_i \iff \tilde{A}_i(t) \geq c_i^*, \qquad c_i^* = \left(\frac{q_i}{\Gamma}\right)^{1/\gamma}. \tag{12}$$

- $q_i \leq \Gamma \Rightarrow c_i^* \leq 1$: adopts at the first influential contact — the rational channel, exactly as in the linear variant (§4.3).
- $q_i > \Gamma \Rightarrow$ needs c_i^* similar adopters; heterogeneity 100% empirical (from q_i), with γ as the single coupling.
- **Degree-dependent immunity**: agent i is out of reach whenever $c_i^* > k_i$ (its degree) — high-aversion, *low-degree* agents can never be socially converted (see §5.1).

What does NOT nest (honesty box).

- Tur's homophily ρ sorts agents by the requirement u *itself*; NDFU's imputation sorts by demographics, and q_i correlates with demographics only partially. The imputed networks are therefore an *empirically disciplined, weaker* version of Tur's preference-sorting. The nesting claim concerns the **adoption rule**; the network is an exogenous input in both models.

One-line summary. With the proximity gate switched off, the γ-form **is** Tur (2024) — same functional form, same γ, with $\xi = \Gamma$ and $u_0 = q$ — on any topology; NDFU's added ingredient is exactly one: the distance gate deciding *which* adopters count in $\tilde{A}_i$.

5 Pros and cons of the two operationalizations

We now have two members of one family on the table:

$$\textbf{Linear-fractional:} \quad \Gamma + \lambda \tilde{E}_i(t) \geq q_i \quad \text{(additive in the adopted } \textit{fraction}\text{)} \tag{13}$$

$$\gamma\textbf{-form (Tur-exact):} \quad \Gamma \cdot \tilde{A}_i(t)^{\gamma} \geq q_i \quad \text{(multiplicative in the adopted } \textit{count}\text{)} \tag{14}$$

This section works through the five questions that decide between them, pedagogically and one at a time, and closes with a verdict table.

5.1 When can an agent never adopt? (immunity, made precise)

In the linear variant, immunity is $\tau_i^* > 1$, i.e.

$$q_i > \Gamma + \lambda. \tag{15}$$

Since $q_i \leq 1$, this region is **non-empty only when** $\lambda < 1 - \Gamma$: with $\lambda \geq 1$ immunity is impossible (a fully-adopted similar neighborhood can compensate any deficit). So the intuition "$q_i, \Gamma \in [0, 1]$ makes this very unlikely" is correct *at large* λ but not in general. Empirically, computed directly from $q_i =$ `mur_score` on the 96 GSS networks (the same 1000 agents in every network, so the shares do not vary across networks; GSS mean $q = 0.585$; isolates 0.11%), share of agents with $q_i > \Gamma + \lambda$:

λ	$\Gamma = 0$	0.1	0.2	0.3	0.4	average over the IUL grid
0.5	66.4%	46.5%	32.2%	16.1%	6.2%	14.0%
0.6	46.5%	32.2%	16.1%	6.2%	0	8.1%
0.7	32.2%	16.1%	6.2%	0	0	4.1%
0.8	16.1%	6.2%	0	0	0	1.7%
0.9	6.2%	0	0	0	0	0.4%
1.0	0	0	0	0	0	0

The share depends only on $\Gamma + \lambda$: 66.4, 46.5, 32.2, 16.1 and 6.2% for $\Gamma + \lambda = 0.5, \ldots, 0.9$, and zero from 1 on.

The immune share is only an **upper bound** on adoption, $\Phi \leq 1 - \Pr(q_i > \Gamma + \lambda)$ – isolates, and with the correct q the dynamics stay far below it at low IUL. At $\lambda = 0.5$ (random seeding) the highest adoption anywhere on the MSP axis is 3–4% for $\Gamma \leq 0.1$ (the seed cluster plus a few), 8% at $\Gamma = 0.2$ (bound 68%) and 27% at $\Gamma = 0.3$ (bound 84%), and jumps to 85–88% at $\Gamma = 0.4$ (bound 94%): most non-immune agents still need a large adopting share of their neighborhood ($\tau_i^* = (q_i - \Gamma)/\lambda$ close to 1), and the cascade stalls before the immune tail binds. The bound is a prediction, not a bug: *when social proof is weak, its reach is bounded.* The current engine cannot produce it because τ is truncated to $[0, 1]$ by construction. (Under the inverted q the immune tail was 10.9% at $\lambda = 0.5$, $\Gamma = 0.2$, and showed as a visible ceiling $\Phi \approx 0.75$.)

In the γ-form, immunity is instead $c_i^* > k_i$, i.e.

$$\left(\frac{q_i}{\Gamma}\right)^{1/\gamma} > k_i \quad \Longleftrightarrow \quad q_i \; > \; \Gamma\, k_i^{\gamma}, \tag{16}$$

which depends on the agent's **degree**: the out-of-reach agents are those both highly averse *and* poorly connected. Arguably more realistic — and a distinctive, testable difference between the two variants.

5.2 The log-space bridge: $\ln a$ vs. $\tilde{E}$

In log space, Tur's reinforcement variable is $\ln a$ — the log of the gated *count*. Is that "just" our normalized exposure? Almost: with $\tilde{E}_i = \tilde{A}_i / k_i$,

$$\ln \tilde{A}_i \; = \; \ln \tilde{E}_i \; + \; \ln k_i. \tag{17}$$

On **degree-regular substrates** — and Tur's Watts–Strogatz networks are degree-homogeneous with $k = 4$ — the term $\ln k_i$ is one global constant, absorbable into Γ (or u_0): identifying the reinforcement with $\ln \tilde{E}$ is then exact up to recalibration. On NDFU's **heterogeneous** networks (degrees roughly 1–80), the two differ by the agent-specific offset $\gamma \ln k_i$:

- using $\tilde{E}$ says "reinforcement is a *share of your attention*";
- using $\tilde{A}$ says "every endorsement adds the same, whether you have 4 neighbors or 40" — at the same adopted *fraction*, a hub receives more reinforcement.

This is the count-vs-fraction fork of §4.6, seen from the logarithm: the analogy $\ln a \leftrightarrow \tilde{E}$ is complete *modulo degree*, and the choice between the two variants is precisely the choice of what to do with $\ln k_i$.

5.3 What γ is — and is not — in Tur 2018 and Tur 2024

Read against both PDFs, the epistemic status of γ is the following.

What their justification does have.

- An **axiomatic characterization** (2018, p. 3; restated in 2024): the updated requirement $f(q; a, \gamma) = q \cdot (1/a)^{\gamma}$ must satisfy (a) decreasing in a for $\gamma > 0$; (b) decreasing in γ for $a > 1$; (c) $\gamma = 0$ replicates the percolation benchmark; (d) $a = 1$ gives $f = q$ (the *first* adopting neighbor only informs; reinforcement starts with the second — a clean echo of complex contagion's "more than one source").
- A **concavity argument** for the restriction $\gamma \leq 1$: "assuming that the marginal effect of each additional neighbor on the total social influence cannot be increasing" (2018), with the reductio that a very large γ trivializes diffusion *on their degree-4 lattices*.
- An interpretable **marginal effect** (2018, Eq. 2): $\Delta q_t^i = -\,q_0^i \gamma \,/\, (a_t^i)^{\gamma+1} \cdot \Delta a$ — the first adopting friends produce the large drops, and the drop is *proportional to the agent's own requirement* (more averse agents receive larger absolute reductions). This proportionality is the seed of the complementarity discussed in §5.5.

What it does not have.

- **No empirical calibration.** In both papers γ is a free "intensity" parameter, introduced with one sentence (2024: "for some innovations, social reinforcement may be more intense than for others … so we introduce social reinforcement as a parameter") and *swept*: $\gamma \in [0, 1]$ continuously in the 2018 phase maps and the 2024 heat maps, with focal values $\{0, 0.5, 1\}$ in 2024's Fig. 5 ("$\gamma = 1$ is maximum social reinforcement"). There are no units, no measurement strategy, no experimental anchor — not even a link to Centola's dose–response evidence, which would have been the natural one.
- **No transportability across degree scales.** All Tur simulations live on $k = 4$ neighborhoods, so $a \in \{1, \ldots, 4\}$ in practice and the reinforcement spans at most a factor $4^{\gamma} \leq 4$. On NDFU's imputed networks (mean degree ≈ 29, hubs beyond 60), the same γ produces reductions up to 29^{γ}: at $\gamma = 1$ a saturated average neighborhood divides the requirement by ≈ 29, a force of a different order of magnitude. Even their $\gamma \leq 1$ bound is reasoned *for* $k = 4$. **Borrowing Tur's γ values would inherit the symbol, not the calibration.**

Verdict. Adopting the γ-form buys (i) exact formal nesting, (ii) the published axioms and the concavity argument for free, and (iii) a citation in the target journal. It does *not* buy empirical values for γ, nor freedom from arguing the coupling's magnitude — if the γ-form is used, γ must be calibrated for *our* degree distribution (e.g. with the same fit-to-current-model procedure used for λ). Epistemically, γ and λ are in the same place: free couplings — with the difference that λ has the stronger microeconomic backing (§5.7).

5.4 Exposure traditions: the power form is wedded to counts

The γ-form cannot be written with normalized exposure, and the reason is structural, not stylistic: the multiplier $\tilde{A}^{\gamma}$ amplifies only because $\tilde{A} \geq 1$. Substituting the fraction $\tilde{E} \in [0, 1]$ gives $\tilde{E}^{\gamma} \leq 1$ — social proof would *reduce* situational utility. There is no natural "fractional γ-form".

Consequences:

- Choosing the γ-form moves the model from the fractional-threshold tradition (Granovetter 1978; Valente 1996; Watts 2002) — the tradition NDFU and `netdiffuseR` live in — to the absolute-count tradition (Centola & Macy 2007). For a *Social Networks* audience this is not neutral.
- **Mitigation 1 (mechanism ≠ measurement):** even if the decision rule runs on counts, Valente-style exposure-at-adoption can still be computed and reported *ex post* as a descriptive statistic (measured thresholds). Nothing forbids reporting $\tilde{E}_i$ at the moment of adoption.

- **Mitigation 2 (tooling):** `netdiffuseR` supports non-normalized exposure (`exposure()` with `normalized = FALSE`), so the count threshold $\lceil (q_i/\Gamma)^{1/\gamma} \rceil$ runs on the same machinery. The loss is conceptual continuity, not implementation.

5.5 Are preferences modified? Composition, not units

Formally, the γ-form does *not* force the "preferences change" reading: $\Gamma \cdot \tilde{A}^{\gamma} \geq q_i$ is symmetric, and the factor $\tilde{A}^{\gamma}$ can be booked on either side —

$$\underbrace{U_i(t) = \Gamma \cdot \tilde{A}_i^{\gamma} \geq q_i}_{\text{situational utility multiplied; preference fixed}} \qquad \Longleftrightarrow \qquad \underbrace{\Gamma \geq q_i \cdot \tilde{A}_i^{-\gamma} = q_{i,t}}_{\text{requirement lowered (Tur's framing)}} \tag{18}$$

with identical behavior. In logs the first reading is a Cobb–Douglas-style utility in two "goods" (product quality, social proof): $\ln U_i = \ln \Gamma + \gamma \ln \tilde{A}_i$. Both sides live on the same $[0, 1]$ utility scale as q_i, so the real distinction is not one of *units* but of **composition**: additive vs. multiplicative.

And the composition carries substance. Compare the marginal value of social proof:

$$\text{linear:} \qquad U_i = \Gamma + \lambda \tilde{E}_i, \quad \frac{\partial U_i}{\partial \tilde{E}_i} = \lambda, \qquad \frac{\partial^2 U_i}{\partial \Gamma \, \partial \tilde{E}_i} = 0 \qquad (\textbf{substitutes}), \tag{19}$$

$$\gamma\text{-form:} \qquad U_i = \Gamma \cdot \tilde{A}_i^{\gamma}, \quad \frac{\partial U_i}{\partial \tilde{A}_i} = \gamma \, \Gamma \, \tilde{A}_i^{\gamma - 1}, \quad \frac{\partial^2 U_i}{\partial \Gamma \, \partial \tilde{A}_i} = \gamma \tilde{A}_i^{\gamma - 1} > 0 \quad (\textbf{complements}). \tag{20}$$

In the linear form, a given (similar) adopting neighborhood contributes the same $\lambda \tilde{E}$ regardless of how good the product is: quality and social proof are *substitutes* (additively separable), and the social contribution needed to convert agent i compensates the utility deficit one-for-one, $\lambda \tilde{E}_i \geq q_i - \Gamma$, independently of Γ's level. In the γ-form the cross-partial is strictly positive (complementarity in the Edgeworth–Pareto sense): *the same adopting neighborhood is worth more for better products.* The mirror statement on the requirement side is Tur 2018's own Eq. 2, where the drop is proportional to q_0^i. The required social proof falls steeply with quality,

$$\ln c_i^* \;=\; \frac{\ln q_i - \ln \Gamma}{\gamma}, \qquad \frac{d \ln c_i^*}{d \ln \Gamma} \;=\; -\frac{1}{\gamma}, \tag{21}$$

an elasticity of $-1/\gamma$ — e.g. at $\gamma = 0.5$, doubling the product's quality divides the required count of similar adopters by four.

So it *can* be framed as preferences-fixed either way, but the multiplicative interaction is a substantive behavioral commitment (quality–social-proof complementarity), not just bookkeeping — and since Tur themselves present the form as requirement-lowering, adopting their exact form and citation invites the "preferences change" reading unless a paragraph defends the other bookkeeping.

The prediction that tells them apart. The social proof needed to convert i depends on the *difference* $q_i - \Gamma$ under the linear form and on the *ratio* q_i/Γ under the γ-form. Because q_i and Γ live on a survey-index scale, the distinction is cardinal: in logs the γ-form is additive too (§4.6). What is scale-free is the shape of the indifference curves between quality and social proof: parallel lines (linear) or hyperbolas that blow up as $\Gamma \to 0$ (γ-form). Which composition is right is a question about the *mechanism* by which peers matter; §5.6 reads the literature on it.

5.6 Substitutes or complements? What the mechanisms imply

Until 2026-11 the only support for the linear form was the modelling convention of Brock & Durlauf (2001). This subsection reads twelve papers against the question of §5.5: does a given adopting neighborhood contribute the same whatever the quality (substitutes), or more for better

products (complements)? The organizing frame is Young (2009, AER 99(5):1899–1924), who sorts diffusion models by *why* prior adoption matters: contagion, social influence (conformity) and social learning. The working hypothesis was: if social proof is a payoff for conforming it adds (substitutes); if it is information about quality it combines multiplicatively (complements), which would suggest substitutes for the collective action of the GSS and complements for the technology of the ATP. The reading confirms the first half and corrects the second.

Conformity and coordination: every model adds, in shares.

- Bass (1969, Mgmt. Sci. 15(5)), pp. 216–217: the hazard of a first purchase is $p + qF(T)$, "a linear function of the number of previous buyers".
- Morris (2000, RES 67(1)), pp. 58, 60; Ellison (1993, Econometrica 61(5)), p. 1049; Blume (1993, GEB 5(3)), p. 391 and Eq. (2.3): each player sums the payoffs of a 2×2 coordination game over her neighbors, so action 1 is a best response when the *proportion* of neighbors playing it exceeds a critical value set by the payoffs. That is the derived threshold (4), obtained rather than assumed.
- Brock & Durlauf (2001, RES 68(2)), Eq. (1), p. 238: private plus social utility plus a shock. The authors adopt the parametric forms because they "render our model both theoretically and econometrically tractable" (p. 238), and their "complementarity" ($\partial^2 S/\partial\omega_i\partial\bar{m} = J$, Eq. 3) is between one's action and the others', not between quality and social proof. NDFU is their model at zero temperature with heterogeneous private utility $h_i = \Gamma - q_i$, a case they allow (§5.3, Eq. 37).
- Soetevent (2006, J. Econ. Surveys 20(2)), pp. 197, 200: the linear form is *assumed* in the empirical literature, and the behavioral motive is almost never identified. P. 222 proposes, as an unexplored extension, interaction weights that depend on the characteristics of both parties, $\gamma_{ij} = \gamma(x_i, x_j)$: NDFU's similarity gate.
- Young (2009), p. 1905: in the social-influence model adoption depends on popularity, "rather than on how good or desirable the innovation has proven to be".

Learning does not yield Tur's product.

- Bikhchandani, Hirshleifer & Welch (1992, JPE 100(5)): agents adopt if $E[V \mid \text{history}] \geq C$ (p. 996). Evidence adds in log-odds — the difference between adoptions and rejections is a sufficient statistic (n. 23, p. 1009) — and it *saturates*: once a cascade starts further adoptions are uninformative, whereas under conformity uniformity "becomes more robust as the number of adopters increases" (p. 1004).
- Young (2009), Eqs. (13), (16), pp. 1911–1912: under learning from outcomes the evidence i needs is her prior deficit divided by the true margin of quality over her cost. There is complementarity, but agents whose cost exceeds the true quality never adopt; Young analyses only $\{i : \mu > c_i\}$ (p. 1912).

Tur's $\Gamma \cdot \tilde{A}^{\gamma}$ cannot be learning: it is unbounded in the count, and learning can never persuade an agent that a product is better than it is. It is a reduced form, with axioms but no mechanism (§5.3).

The experiments establish the channel, not the functional form. Salganik, Dodds & Watts (2006, Science 311) show that a social signal makes success more unequal and the quality–success mapping more convex (Fig. 3), but this is an aggregate market-share result. Krumme et al. (2012, PLoS ONE 7(5)), p. 2, separate sampling (where the social signal and appeal *add*) from downloading (driven by quality alone); the product is *between* stages, which NDFU already has as its awareness condition. Cai, Chen & Fang (2009, AER 99(3)) find that a top-five ranking raises

demand by 13–20% (Tables 3, 5), more so among infrequent customers (Table 8), and cannot separate learning from conformity (n. 8, p. 865). Tucker & Zhang (2011, Mgmt. Sci. 57(5)) find that popularity helps narrow-appeal vendors more, with an additive utility (p. 829). **None of them estimates the sign of $\partial^2 U/\partial\Gamma\,\partial\tilde{E}$ at the individual level.** (Manski 1993 is cited only through Brock & Durlauf and Soetevent.)

Where quality lives (our derivation from Morris's game). In a two-action game with each neighbor, the advantage of adopting is always $-a + (a+b)\,\tilde{E}$, with a the loss from adopting when the neighbor did not and b the gain when she did. It is linear in the share. If quality is value that does not need others, it lowers a and shifts the line: *substitutes*. If quality is value realized only jointly (a network good, a protest that needs a crowd), it raises b and tilts the line: *complements, bounded and in shares*, $\Gamma + (\lambda_0 + \varphi\Gamma)\tilde{E} \geq q$, not Tur's form.

Four boxes.

Box	Why peers matter	Composition ($\partial^2 F/\partial\Gamma\partial\tilde{E}$)	Status in NDFU
1. Conformity	legitimacy, safety in numbers, fashion, pure coordination: utility that does not depend on quality	sum, $\Gamma + \lambda\tilde{E}$ (0)	main rule
2. Joint value	the behavior pays only if others adopt too	bounded product in shares, $\Gamma + (\lambda_0 + \varphi\Gamma)\tilde{E}$ (φ)	robustness, level 1
3. Learning	peers inform about an uncertain quality	evidence that adds and saturates, capped by true quality	out of scope
4. Tur	none (reduced form)	unbounded product in counts, $\Gamma \cdot \tilde{A}^{\gamma}$ ($\gamma\tilde{A}^{\gamma-1}$)	robustness, level 2

The exposure axis of §8 is a second, independent knob: *how* adopters are counted into $\tilde{E}_i$ before being combined with Γ. Its two arms have natural mechanisms too: the independent reinforcement θ (redundant sources discounted) is the logic of box 3, and the cohesive reinforcement κ (pairs of adopters who know each other) the logic of boxes 1–2.

By survey. GSS (collective action): legitimacy and safety are box 1, the efficacy of a protest that needs a crowd is box 2; the sum is the natural rule. ATP (technology): learning (box 3), network effects (box 2) and fashion (box 1); the sum is a reasonable simplification, and box 3 is a stated limitation — technology adoption is the textbook case of learning (Ryan and Gross's hybrid corn, the running example of Young 2009).

The defense of the additive rule. Quality and social proof add because peers contribute a utility of their own that does not depend on quality (legitimacy, safety, pure coordination). If that utility grows with quality (box 2), the result holds. Pure learning (box 3) is outside the scope of the model. Two reasons beyond the convention: (i) it is the best-response form of every local-interaction game; (ii) it is what the model needs, because everything interesting happens to agents with $q_i > \Gamma$ who adopt through their peers — if peers only informed about a common quality, those agents would learn not to adopt.

Robustness. The premium does not depend on the composition. The numbers come from the script `06f_rule_robustness.R` (about 2.5 min), with base exposure and the networks, twins, seeds and frozen gate of 05. Premium = mean over the $\Gamma \times h$ plane of adoption in the network minus its DP twin, in pp, with 95% run-bootstrap intervals ($B = 2000$); random seeding.

- **Level 1, box 2, no new simulation.** For each Γ the rule $\Gamma + (\lambda_0 + \varphi\Gamma)\tilde{E} \geq q$ *is* the main rule with $\lambda(\Gamma) = \lambda_0 + \varphi\Gamma$: a diagonal path through the (Γ, λ) grid already swept, interpolated linearly in λ and clamped to $[0.5, 1]$ (a pure network good, $\lambda_0 = 0$, is out of range).
- **Level 2, box 4, new sweep.** $\Gamma \cdot \tilde{A}^{\gamma} \geq q$ is one more vector of node thresholds, $\tilde{E}_i \geq (q_i/\Gamma)^{1/\gamma}/k_i$, so the engine runs it unchanged. Checks: the trick reproduces the main rule cell by cell, the engine equals an independent R cascade by counts in 2016/2016 cells, and six hand-solved cases hold. $\gamma \in \{0.2, 0.25, 1/3, 0.5, 0.75, 1\}$, chosen for degree ≈ 30 (Tur's values are for degree 4); $\Gamma \in [0.025, 1]$ ($\Gamma = 0$ is undefined).

Level 1, $\Gamma + \lambda(\Gamma)\,\tilde{E} \geq q$

$\lambda(\Gamma)$	GSS ×2	ATP ×2	GSS ×1	ATP ×1
$\lambda = 0.5$	5.0 [4.8, 5.2]	5.5 [5.1, 5.8]	5.5 [5.3, 5.6]	5.8 [5.4, 6.2]
$\lambda = 0.6$	5.6 [5.4, 5.8]	6.0 [5.6, 6.4]	6.1 [5.9, 6.3]	6.4 [6.0, 6.7]
$\lambda = 0.7$	6.4 [6.2, 6.6]	6.5 [6.1, 6.9]	6.9 [6.7, 7.2]	6.9 [6.5, 7.3]
$\lambda = 0.8$	7.0 [6.8, 7.2]	7.1 [6.7, 7.5]	7.6 [7.3, 7.9]	7.4 [7.0, 7.8]
$\lambda = 0.9$	7.5 [7.3, 7.7]	7.6 [7.2, 8.0]	8.2 [7.8, 8.4]	7.9 [7.5, 8.4]
$\lambda = 1.0$	8.1 [7.8, 8.3]	8.1 [7.6, 8.5]	8.5 [8.2, 8.9]	8.4 [7.9, 8.9]
$0.5 + 0.25\,\Gamma$	5.5 [5.3, 5.6]	5.9 [5.5, 6.3]	5.9 [5.7, 6.1]	6.3 [5.8, 6.6]
$0.5 + 0.5\,\Gamma$	5.8 [5.7, 6.0]	6.2 [5.8, 6.6]	6.3 [6.1, 6.5]	6.6 [6.2, 6.9]
$0.5 + \Gamma$	6.4 [6.2, 6.6]	6.7 [6.3, 7.1]	7.0 [6.7, 7.1]	7.1 [6.6, 7.4]
$0.5 + 2\,\Gamma$	7.2 [7.0, 7.4]	7.4 [7.0, 7.7]	7.8 [7.5, 8.1]	7.7 [7.2, 8.0]
$1 - 0.5\,\Gamma$	7.8 [7.5, 8.0]	7.8 [7.4, 8.2]	8.4 [8.1, 8.7]	8.2 [7.7, 8.6]

Level 2, $\Gamma \cdot \tilde{A}^{\gamma} \geq q$

γ	GSS ×2	ATP ×2	GSS ×1	ATP ×1
0.20	4.6 [4.5, 4.8]	5.1 [4.7, 5.5]	5.0 [4.8, 5.2]	5.5 [5.0, 5.8]
0.25	5.0 [4.8, 5.1]	5.5 [5.0, 5.9]	5.4 [5.2, 5.6]	5.8 [5.4, 6.2]
0.33	5.4 [5.2, 5.6]	6.0 [5.5, 6.4]	5.9 [5.7, 6.1]	6.3 [5.8, 6.7]
0.50	5.9 [5.6, 6.2]	6.5 [6.0, 7.0]	6.5 [6.2, 6.8]	6.9 [6.3, 7.4]
0.75	7.0 [6.4, 7.5]	6.9 [6.3, 7.4]	7.4 [6.9, 7.9]	7.3 [6.6, 7.8]
1.00	7.0 [6.3, 7.6]	7.3 [6.5, 8.0]	7.4 [6.9, 7.9]	7.5 [6.6, 8.2]

The premium is positive everywhere: 4.6–8.5 pp on the $\lambda(\Gamma)$ paths over all four sets and five seedings (lowest lower bound 4.4), and 4.0–8.8 pp under the γ-form (lowest lower bound 3.9; positive in all 24 networks in 118 of 120 set × seeding × γ combinations, in 23 in the other two). As with the linear rule, a region of the plane favors the DP twin (4–17% of the cells in GSS ×2, by γ) without changing the sign of the average. At low γ many agents are immune (29% at $\gamma = 0.2$, 1% at $\gamma = 1$), and immunity depends on degree (§5.1). Not yet run: boxes 2 or 4 crossed with θ or κ.

5.7 The Brock–Durlauf microfoundation of the linear form

The linear variant is the deterministic limit of the canonical discrete-choice-with-social-interactions model of Brock & Durlauf (2001, *Review of Economic Studies* 68(2):235–260). In their setup, agent i chooses a binary action and derives utility

$$V_i(\omega_i) \;=\; \underbrace{h_i(\omega_i)}_{\text{private utility}} \;+\; \underbrace{J\,\omega_i\,\bar{m}_i^{e}}_{\text{social utility (conformity)}} \;+\; \underbrace{\varepsilon_i(\omega_i)}_{\text{random shock}}\;, \tag{22}$$

where $\bar{m}_i^{e}$ is the expected mean choice of i's *reference group* and $J \geq 0$ is the social-interaction (conformity) coefficient. Three points of contact:

1. **Additive separability of private and social utility is the standard assumption** of that literature — exactly the structure of the linear rule — adopted for tractability (p. 238); the substantive reasons for it are in §5.6. Mapping: the private net utility of adopting is $\Gamma - q_i$; the social term is $\lambda \tilde{E}_i$, with $\tilde{E}_i$ playing $\bar{m}_i^e$ (the adoption rate of the reference group) and λ playing J. Adoption when total utility is non-negative: $\Gamma - q_i + \lambda \tilde{E}_i \geq 0$ is (3) rearranged.

2. **NDFU's additions are well-defined deviations, not violations:** the reference group is not exogenous but *similarity-gated* (the $\sigma(h - d_{ij})$ filter selects who belongs to i's effective reference group — our contribution), and the shock is degenerate ($\varepsilon \to 0$), turning Brock–Durlauf's logit choice probabilities into a deterministic threshold rule. Reintroducing a small ε would give a smooth (logit) adoption rule — a natural robustness variant.

3. **J has an econometric life that γ lacks.** The Brock–Durlauf coefficient underpins the social-multiplier / social-interactions estimation literature (see Blume, Brock, Durlauf & Ioannides 2011, *Handbook of Social Economics*, for identification), so λ inherits a path toward empirical anchoring. As a bonus, Brock & Durlauf themselves develop the mean-field correspondence of (22) with the Ising model — which connects this microfoundation directly to NDFU's criticality analysis.

5.8 Verdict table and recommendation

	Linear-fractional — $\Gamma + \lambda \tilde{E}_i \geq q_i$	**γ-form (count)** — $\Gamma \cdot \tilde{A}_i^{\gamma} \geq q_i$
Answer to Bianchi	✓ equally bottom-up (threshold derived from measured q)	✓ equally bottom-up
Backing of the coupling	λ: best-response form of local-interaction games when peers give utility of their own (§5.6); Brock–Durlauf microfoundation (§5.7); its magnitude still needs arguing	γ with Tur precedent (axioms + concavity), but **no empirical calibration and no transportability** across degree scales (§5.3)
Nesting in Tur 2024	only qualitative	**exact, on any graph** (gate off; ξ=Γ, u_0=q, same γ)
Exposure tradition	✓ Valente/Granovetter fractional; `netdiffuseR` native	counts (Centola–Macy/Watts); Valente exposure only as ex-post reporting (§5.4)
Preferences-fixed framing	✓ clean ("situational utility rises"; Brock–Durlauf)	defensible, but against the grain of the very citation it borrows (§5.5)
Quality × social proof	substitutes (social contribution independent of Γ): conformity, safety, pure coordination (box 1)	**complements** ($\partial^2 U / \partial \Gamma \, \partial \tilde{A} > 0$), unbounded; no mechanism yields it, not even learning (box 4, §5.6)
Immunity	$q_i > \Gamma + \lambda$, degree-independent (§5.1)	$q_i > \Gamma k_i^{\gamma}$, **degree-dependent** (averse + poorly connected) — arguably more realistic
Empirical status	✓ **run on the full design** (§7.1); λ swept over $[0.5, 1.0]$, not calibrated	**run as robustness** (`06f`): premium 4.0–8.8 pp, positive everywhere (§5.6)
Hub behavior	bounded ($\tilde{E} \leq 1$)	at high γ, a saturated hub receives reinforcement $\sim k^{\gamma}$ — may over-fire on dense networks

Recommendation (updated 2026-11-10). Use the **linear-fractional variant as the primary operationalization**, defended by mechanism, not by convention (§5.6): peers contribute a utility of their own that does not depend on quality; if it grows with quality (box 2) the result holds; pure learning (box 3) is outside the scope of the model. Keep the **γ-form as the formal bridge to Tur and as a robustness check**: the exact nesting (§4.6) in the text, and the γ

sweep of `06f` — with γ chosen for our degree scale, never Tur's numbers — in the supplement, next to the $\lambda(\Gamma)$ paths. Both done: the conclusions do not hinge on the additive-vs-multiplicative composition.

6 Side-by-side with the current model and Tur

	NDFU current	Tur 2024	**Unified rule (both variants)**
Decision rule	two channels (OR)	single utility rule	**single utility rule**
Threshold source	exogenous $\mathcal{N}(\tau_{\text{mean}}, \tau_{\text{sd}})$	emerges from reinforcement	**derived from measured** q_i: $(q_i - \Gamma)/\lambda$ or $(q_i/\Gamma)^{1/\gamma}$
Preferences modified during diffusion?	no	yes (u lowered)	**no** under the situational-utility framing (§4.2, §5.5)
Distance enters via	stochastic gate in $\tilde{E}$	— (absent)	**same stochastic gate**, in $\tilde{E}$ or in $\tilde{A}$
Heterogeneity source	q empirical + τ artificial	u artificial	q **empirical only**
Free params (beyond Γ, h)	τ_{mean}, τ_{sd}	γ	λ (linear) or γ (γ-form)
Nests Tur 2024?	no	—	**exactly via the γ-form** (§4.6); linear form only qualitatively
`netdiffuseR` as-is?	yes	no	**yes** (fractional or count `threshold.dist`)
Main risk	Bianchi critique	distance-blind	results depend on the free λ (reported as functions of λ, §7)

7 Empirical status: the coupling λ and the premium

- λ **is a free coupling, swept, not calibrated.** The study sweeps $\lambda \in \{0.5, 0.6, \ldots, 1.0\}$ and reports every result as a function of λ (§7.1). Under the linear-fractional variant $\tau_i^* = \max\{0, (q_i - \Gamma)/\lambda\}$, so the implied threshold distribution depends on Γ, λ and the empirical q distribution.
- **Why this range.** $\lambda = 1$ is the **immunity boundary**: since $q \leq 1$, $\lambda \geq 1$ leaves no structurally immune agent at any $\Gamma > 0$, while $\lambda < 1$ leaves an immune share $\Pr(q > \Gamma + \lambda)$ (§5.1). At $\lambda = 0.5$ two thirds of the agents are already immune at $\Gamma = 0$. There is no theoretical bound at $\lambda = 1$ (λ is a utility contribution, like Brock–Durlauf's unbounded J).
- **Retired: the calibration against the legacy engine.** An earlier sweep over λ from 0.3 to 1.5 (random seeding) found a best fit to the legacy surface ($\tau \sim \mathcal{N}(0.40, 0.12)$, $\beta_{\text{DP}} = -0.850$) at $\lambda \approx 0.7$. Both engines read the propensity score as q (the inverted-q bug), so it compared two invalid surfaces, and the study does not need a calibrated λ. Neither $\lambda \approx 0.7$ nor -0.850 should be cited (retired 2026-11-02).
- The mirror γ-form sweep (count threshold $(q_i/\Gamma)^{1/\gamma}$ on the same machinery) was run on 2026-11-10 (`06f`, §5.6). The ATP/innovation system is swept since PR #4 (§8).

7.1 The structural premium survives the unified rule

Record of the previous design. This subsection reports the 96 ERGM networks of 02 (the same 1000 agents) with the *annealed* gate. Since PR #4 the sweep uses calibrated networks (x2 main, x1) and the frozen gate: see §8, which also compares the two.

Run by `scripts/05_unified_diffusion_sweep.R`: GSS vs. genuine degree-preserving twins (`igraph::rewire(keeping_degseq)`, built in memory with the recipe of

`scripts/04_make_DP_networks.R`), $\lambda \in \{0.5, \ldots, 1.0\}$, 41 IUL levels $\times$ a 25-level MSP grid (step 1/24, nesting the legacy 13-level grid), 96 networks, five seeding strategies (1,180,800 simulations each), with q_i = `mur_score` = 1 − propensity$_i$ (GSS mean 0.585). Analysis in `scripts/06_premium_sensitivity.R` (λ profile, link-free diagnostics) and `scripts/06b_premium_bootstrap.R` (joint BAM of the strategies and run-level bootstrap; all intervals below).

Seed matching. The primary seed is drawn once per run, independently of λ. For *random*, *marginal* and *central* it is the *same node* in a GSS network and its DP twin; *closeness* and *eigen* use each topology's own most central node. The seed cluster is the primary plus its nearest nodes, 0.40$\times$ the network's maximum degree in total ($\approx$ 28 seeds). A network and its twin share the degree sequence, so the cluster has the same size in both (checked in all 96 runs of every strategy), and the strategies differ only in where they seed. With a low-degree primary the cluster reaches into the second neighborhood, so its shape differs too, not only its center. The *random* strategy is the reference.

Uncertainty: the run is the unit. The 1025 cells of a run share the GSS network, its DP twin and the primary draw, so they are not independent observations: the unit of replication is the **run** (96 per topology). Run k also uses the same network, twin and primary draw in every strategy, so contrasts between strategies are **paired**. All intervals come from a run-level bootstrap: resample `run_id` (1..96) with replacement, with one set of weights shared by GSS, DP and every strategy, and refit ($B = 1000$; 95% percentile intervals). The model, per λ, is a joint BAM

$$\operatorname{logit}\Phi \;=\; \alpha_s + \beta_{\mathrm{DP},s}\,\mathbf{1}_{\mathrm{DP}} + f_s(\Gamma, h), \qquad f_s = \texttt{te}(\Gamma, h, k = 12 \times 12) \text{ per strategy } s,$$

fitted on run-aggregated counts (exact: the model has no run term). Three choices, decided by checking them against the data:

- **The surface varies by strategy.** With one shared surface, each strategy's misfit leaks into the strategy $\times$ topology terms and flattens the contrasts (at $\lambda = 1$, central − random shrinks from -0.21 to -0.08). With strategy-specific surfaces the joint model factorizes and reproduces the separate per-strategy fits exactly; what it adds is the paired bootstrap of the contrasts.
- **It does not vary by topology.** β_{DP} is *defined* as the constant logit shift between GSS and DP. A topology-specific surface would absorb the premium itself. The premium is really a shift of the tipping frontier, and its shape is described by the link-free gap and the MSP_c shift, not by the BAM.
- **Basis 12×12, not $k = 10$.** The $s(\Gamma, h, k = 10)$ surface of 05/06 is too stiff for the sharp frontier (deviance explained 0.94–0.99). β_{DP} converges at a 12×12 tensor basis; $k = 10$ understates $|\beta|$ by 3–13%, more at high λ. Contrasts between strategies barely move. $k = 10$ values are kept for continuity.

The pooled premium is the unweighted mean of the six per-λ estimates ("λ-mean"). The pooled fit of 05 has one intercept per λ but puts one surface under all of them, and is attenuated (random: -0.718 vs. -0.763 with the same $k = 10$ basis; an earlier version also left out the λ intercepts and gave -0.684).

λ	β_{DP} (odds) te 12×12	odds drop	β_{DP}, $k = 10$ (as in 05)	$\bar{\Phi}_{\mathrm{GSS}} - \bar{\Phi}_{\mathrm{DP}}$ (pp)	% of plane with gap > 0.10	mean gap in that band (pp)
0.5	-0.499 $[-0.540, -0.434]$	-39%	-0.482	3.98 [3.86, 4.05]	18.1 [17.6, 18.4]	16.5 [16.2, 16.7]
0.6	-0.630 $[-0.692, -0.536]$	-47%	-0.589	4.52 [4.39, 4.60]	20.3 [19.0, 20.7]	17.6 [17.3, 18.0]
0.7	-0.834 $[-0.934, -0.689]$	-57%	-0.753	5.46 [5.29, 5.58]	20.8 [20.3, 21.1]	21.3 [20.9, 21.7]
0.8	-0.971 $[-1.101, -0.787]$	-62%	-0.851	5.95 [5.78, 6.08]	20.8 [20.6, 21.1]	23.4 [22.8, 23.9]
0.9	-1.054 $[-1.216, -0.837]$	-65%	-0.922	6.30 [6.11, 6.44]	21.2 [20.8, 22.4]	24.6 [23.8, 25.2]
1.0	-1.118 $[-1.306, -0.881]$	-67%	-0.980	6.57 [6.37, 6.72]	21.5 [21.1, 21.8]	25.8 [25.2, 26.3]
mean	-0.851 $[-0.963, -0.697]$	-57%	-0.763	5.46 [5.30, 5.57]	20.4 [20.0, 20.8]	21.5 [21.1, 21.9]

Random seeding; 95% run-bootstrap intervals below each estimate (the last two columns are the band statistics of 06). MSP_c, GSS $\to$ DP: .085$\to$.156, .088$\to$.157, .082$\to$.153, .079$\to$.151, .078$\to$.147 and .077$\to$.148 for $\lambda = 0.5, \ldots, 1.0$; the shift, .071 [.065, .073], .069 [.063, .075], .071 [.065, .072], .071 [.065, .073], .069 [.062, .074] and .071 [.066, .074].

Reading. The premium survives the bottom-up engine across the whole λ band and for all five seeding strategies, and no interval comes near zero: rewiring the ties lowers the odds of adoption by 39–67% under random seeding (λ-mean $\beta_{\mathrm{DP}} = -0.851$ $[-0.963, -0.697]$, -57%; probability scale 5.46 pp $[5.30, 5.57]$). It grows smoothly and monotonically with λ (steps per decile -0.131, -0.203, -0.137, -0.083, -0.063): the stronger social proof is, the more the real fabric matters. The honest intervals are 7–18 times the naive BAM standard errors for the hub strategies and 64–220 times for random and marginal, whose primary seed is a lottery.

λ is a free coupling, not a calibrated constant, so the paper should report results as functions of λ. Two scales must be read together:

- on the **odds** scale the premium more than doubles over the band ($-0.50 \to -1.12$; $-0.48 \to -0.98$ with the $k = 10$ surface);
- on the **probability** scale it grows by +65% (4.0 $\to$ 6.6 pp), with much tighter intervals and no dependence on the surface basis.

Part of the difference is a link effect (cells with $\Phi > 0.95$ go from 40% to 61% of the plane for GSS, from 32% to 48% for DP, and the logit inflates a given probability gap there), but most of the growth is genuine structural separation.

The λ-robust statements (recommended for the paper):

1. **Where it lives.** The premium is concentrated in an L-shaped band along the tipping frontier: 18–22% of the (IUL, MSP) plane, where GSS gains 16–26 percentage points (λ-mean 20.4% $[20.0, 20.8]$ and 21.5 pp $[21.1, 21.9]$; no per-λ interval is wider than ± 1.5). Outside that band the two topologies agree (both dead, or both saturated). Structure decides where the system is undecided.
2. **Tipping shift.** DP needs 1.8–1.9 times the social openness to tip ($MSP_c \approx 0.077$–0.088 for GSS vs. 0.147–0.157 for DP; ratio λ-mean 1.87 $[1.81, 1.89]$). The absolute shift is flat in λ: 0.069–0.071 under random seeding, λ-mean 0.070 $[0.065, 0.073]$. Its per-λ intervals are about ± 0.005 wide, so the flatness is well inside the noise; MSP_c moves on the 1/24 grid, so these intervals are coarse. The five strategies cannot be told apart on this scale (λ-mean from 0.065 $[0.064, 0.070]$ for central to 0.071 $[0.065, 0.074]$ for marginal).

3. **Channel split** ($k = 10$ surface, as in 06). The rational channel is flat in λ ($\beta = -0.146$ $[-0.152, -0.138]$ at every λ; 1.81 pp $[1.75, 1.86]$) while the social channel carries the growth (-0.206 $[-0.208, -0.202] \to -0.372$ $[-0.384, -0.357]$; 2.17 $\to$ 4.76 pp). The same holds in all five strategies (rational -0.140 to -0.154). The rational channel is *not* topology-blind: its adopters still need one gate-passing contact, and DP ties are socially more distant (mean Gower 0.229 $\to$ 0.325), so the gate rejects more often.

Seeding strategies. λ-mean and endpoint β_{DP} (te 12×12), the λ-mean probability-scale gap and the tipping shift, by strategy (mean seed-cluster size in parentheses; 95% run-bootstrap intervals):

strategy	β_{DP} λ-mean	β_{DP} $\lambda = 0.5$	β_{DP} $\lambda = 1.0$	$k = 10$ λ-mean	gap (pp) λ-mean	MSP_c shift λ-mean
central (28.7)	-0.965 $[-0.976, -0.954]$	-0.533 $[-0.540, -0.526]$	-1.325 $[-1.344, -1.304]$	-0.877	5.61 [5.55, 5.68]	.065 [.064, .070]
closeness (28.7)	-0.961 $[-0.973, -0.951]$	-0.535 $[-0.542, -0.529]$	-1.320 $[-1.340, -1.299]$	-0.874	5.60 [5.54, 5.66]	.067 [.065, .071]
eigen (28.7)	-0.949 $[-0.961, -0.936]$	-0.523 $[-0.529, -0.516]$	-1.319 $[-1.343, -1.295]$	-0.864	5.55 [5.47, 5.62]	.068 [.064, .071]
random (28.4)	-0.851 $[-0.963, -0.697]$	-0.499 $[-0.540, -0.434]$	-1.118 $[-1.306, -0.881]$	-0.763	5.46 [5.30, 5.57]	.070 [.065, .073]
marginal (28.4)	-0.852 $[-0.965, -0.700]$	-0.508 $[-0.549, -0.445]$	-1.091 $[-1.282, -0.865]$	-0.762	5.44 [5.29, 5.54]	.071 [.065, .074]

Paired contrasts (same run in both strategies). On β, a negative value means the first strategy has the *larger* premium; on the gap, a positive one does:

contrast	$\Delta\beta_{\mathrm{DP}}$, λ-mean	$\lambda = 0.5$	$\lambda = 1.0$	Δ gap (pp), λ-mean
central − random	-0.114 $[-0.271, -0.002]$	-0.034 $[-0.098, 0.008]$	-0.207 $[-0.448, -0.014]$	0.15 [0.05, 0.31]
closeness − random	-0.110 $[-0.267, 0.003]$	-0.036 $[-0.101, 0.005]$	-0.202 $[-0.440, -0.009]$	0.14 [0.03, 0.30]
eigen − random	-0.098 $[-0.252, 0.017]$	-0.024 $[-0.089, 0.018]$	-0.201 $[-0.441, -0.006]$	0.08 [−0.04, 0.24]
random − marginal	0.001 $[-0.246, 0.249]$	0.009 $[-0.086, 0.106]$	-0.026 $[-0.404, 0.362]$	0.02 [−0.15, 0.20]
central − closeness	-0.004 $[-0.013, 0.006]$	0.002 $[-0.003, 0.008]$	-0.005 $[-0.022, 0.011]$	0.01 [−0.04, 0.07]
central − eigen	-0.016 $[-0.026, -0.005]$	-0.010 $[-0.017, -0.004]$	-0.006 $[-0.026, 0.012]$	0.07 [0.01, 0.13]
closeness − eigen	-0.012 $[-0.023, -0.002]$	-0.012 $[-0.019, -0.006]$	-0.001 $[-0.022, 0.021]$	0.06 [−0.01, 0.12]

- **The three hub strategies are interchangeable.** Central and closeness do not differ ($\Delta\beta = -0.004$ $[-0.013, 0.006]$); eigen's premium is distinguishably smaller, but by 1.6% of β (0.07 pp). The hub premium is also the most precisely estimated (interval width ≈ 0.02 in β): the primary is fixed by the network, so runs differ only in network and dynamics.
- **Random and marginal coincide** ($\Delta\beta = 0.001$ $[-0.246, 0.249]$; gap 0.02 pp $[-0.15, 0.20]$): a typical and a peripheral primary seed equally well. Their premium is ten times less precise than the hubs' (width ≈ 0.27): where the primary lands is a lottery.
- **The hubs add a small premium, borderline on the odds scale.** Central − random is -0.114 $[-0.271, -0.002]$ on the λ-mean and 0.15 pp $[0.05, 0.31]$ on the gap (3% of the 5.5-pp premium; t interval $[0.02, 0.28]$). It grows with λ: indistinguishable from zero at $\lambda = 0.5$ and 0.7–0.8, distinguishable at $\lambda \geq 0.9$ (-0.207 $[-0.448, -0.014]$ at $\lambda = 1$; 0.31 pp

$[0.12, 0.52]$) and, barely, at $\lambda = 0.6$. Closeness behaves like central (-0.110 $[-0.267, 0.003]$; 0.14 pp $[0.03, 0.30]$). Eigen's margin is the smallest and not distinguishable on the λ-mean (-0.098 $[-0.252, 0.017]$; 0.08 pp $[-0.04, 0.24]$), but at $\lambda = 1$ all three hubs are ($\Delta\beta \approx -0.20$; 0.29–0.33 pp).

A reverse-premium corner (conditional, not universal). In a small region — high openness ($h \gtrsim 0.6$, median 0.83) and moderate attractiveness ($\Gamma \approx 0.275$–0.45) — **DP beats GSS**. To tell it from noise, each cell gets the paired z of its gap over the 96 runs, and counts as a **robust DP win** only if it clears a one-sided *simultaneous* threshold: max-T over the 1025 cells of its strategy $\times$ λ panel, familywise 5%, from the same run bootstrap ($z < -3.6$ to -3.8, against -2 pointwise; `premium_reverse_corner.csv`):

- **Robust at $\lambda = 0.5$–0.7, identically in all five strategies.** 47–48, 59–61 and 38 corner cells clear the threshold, including *every* cell where DP wins by more than 2 pp (2.6–3.4%, 2.4–2.6% and 0.8% of the plane, up to 7.0, 5.8 and 3.0 pp). At $\lambda = 0.5$ it spills just past the box ($\Gamma = 0.475$–0.525, ≤ 1 pp).
- **Detectable but negligible at $\lambda = 0.8$** (12–13 corner cells, ≤ 0.14 pp, plus 8 cells at $\Gamma = 0.25$ with ≤ 0.35 pp); at $\lambda \geq 0.9$ at most 0.04 pp remains.
- **The low-IUL strip** ($\Gamma \leq 0.1$, DP ahead by ≤ 0.05 pp) is noise for the hub strategies and random: no cell clears the threshold (up to 19 pass pointwise). For marginal it is systematic (30–48 cells per λ, $h \gtrsim 0.5$) but negligible (≤ 0.14 pp): an effect of the seed cluster's geometry, not a reversal of the premium.

The corner tracks the immunity boundary. Its strong part (> 2 pp) lies at $\Gamma + \lambda \approx 0.85$–$0.975$, where a small immune tail remains (13 to 1% of agents), and the robust cells drift to lower Γ as λ grows ($\Gamma = 0.35$–0.45 at $\lambda = 0.5$, 0.275–0.375 at 0.7, 0.275–0.3 at 0.8). There the most averse non-immune agents need nearly all of their neighbors to adopt (τ_i^* close to 1). The mechanism is Tur (2024)'s own: when the gate is wide open, homophily *isolates the resistant among the resistant*, whereas random mixing seats adopters next to reluctant agents and converts them. Once influence stops being similarity-gated, the sign of the structural effect can flip (the similarity-vs-diversity question). This is a lead, not an established mechanism (it did not hold under the inverted q). The corner keeps the premium claim properly conditional, but it is a weak-coupling feature and should not be presented as present at every λ.

Immunity under the correct q. Agents with $q_i > \Gamma + \lambda$ can never adopt. At $\Gamma = 0$ they are 66, 47, 32, 16, 6 and 0% of agents for $\lambda = 0.5, \ldots, 1.0$; averaged over the IUL grid, 14.0, 8.1, 4.1, 1.7, 0.4 and 0%. Mean adoption over the plane is correspondingly lower (random seeding: $\bar{\Phi}_{\text{GSS}} = 0.61$–$0.74$, $\bar{\Phi}_{\text{DP}} = 0.57$–$0.68$).

What changed with respect to the inverted q. Until 2026-10-05 the engine read the propensity score itself as q_i (high = *more* willing), the opposite of its role as a minimum utility requirement (GSS mean 0.415 instead of 0.585); the same inversion affected the legacy engine. With the correct q: (i) the premium survives in every strategy and keeps its shape (L-shaped band, flat rational channel, stable tipping shift of ≈ 0.065–0.07); (ii) it is smaller on the odds scale (λ-mean with $k = 10$: central $-1.07 \to -0.88$, random $-0.81 \to -0.76$) and, relative to its level, steeper in λ (β at $\lambda = 1$ over $\lambda = 0.5$: central $2.0 \to 2.3$, random $1.6 \to 2.0$); (iii) the reverse corner moves to higher Γ (from 0.10–0.275, by about the 0.17 shift in mean q) and is no longer present at every λ; (iv) the immune share rises 2.7-fold or more (IUL-grid average $5.2 \to 14.0\%$ at $\lambda = 0.5$). The inverted-q outputs are archived in `output/**/_qinverted/` (closeness and eigen references: the pre-seed-fix runs in `_prefix/`, which match the seed-matched code for the deterministic strategies: central -0.926 vs. -0.928). Those runs predate the common seed budget: except for central,

whose cluster is the same under both rules, they seeded 0.40× the degree of the primary, so central is the like-for-like comparison.

Still open.

- **What the intervals cover.** The bootstrap treats the 96 networks as exchangeable draws from the fitted ERGM. They share the same 1000 imputed agents (the q composition is fixed), so the intervals cover tie formation, the seed draw and the dynamics, not survey sampling of the agents or uncertainty in the ERGM coefficients.
- **Specification.** The level of β_{DP} depends on the surface basis (3–13% between $k = 10$ and te 12×12). That is not in the bootstrap intervals; the contrasts are robust to it, and the link-free gap, which does not depend on it, should lead in the paper.

8 PR #4: calibrated networks, frozen gate and the exposure axis

The results of §7.1 were computed on the 96 ERGM networks of `02_GSS_network_ergm.R` with the annealed gate. Branch `rcpp-engine` (PR #4, stacked on PR #3) changes three things and keeps the rest (the rule, $q =$ `mur_score`, the 41×25 grid, $\lambda \in \{0.5, \ldots, 1.0\}$, the five seeding strategies and the common seed budget):

1. **the networks**: calibrated by race to the GSS 2004 ego networks (level x1), then rewired to twice the triangles of every node (level x2, the main networks);
2. **the gate**: frozen (one coin per tie and simulation) instead of annealed;
3. **the exposure**: the main text keeps the base exposure $E_i = a_i/d_i$; a robustness axis with one formula and two dials goes to the supplement.

The same recipe builds networks for the GSS 2004 sample (collective action) and for the ATP 2014 sample (new products). ATP has no ego networks, so its networks use the GSS 2004 targets, as the ERGM of 02 used the GSS-based coefficients of Smith et al. (2014).

8.1 Networks: calibrated by race, then twice the triangles of every node

What was wrong with the ERGM networks. The ERGM of 02 is dyad independent with one race coefficient and no degree term. Its degree by race is an artifact (whites 33.9, blacks 16.8, Hispanics 10.0, Asians 7.5 contacts, against GSS 2004 ratios of confidants $1.09 : 0.68 : 0.90 : 0.84$), it has almost no closure (transitivity 0.065, about twice the density), and its 100 networks share the same 1000 agents.

Level x1 (`scripts/02_network_calibrated.R`). For every network the 1000 nodes are resampled with replacement from the survey's complete cases. Ties are independent coins with

$$\operatorname{logit} p_{ij} = \delta_{r(i)} + \delta_{r(j)} + \gamma_r \, \mathbf{1}[r(i) = r(j) = r] + \beta^\top x_{ij},$$

where $\beta^\top x_{ij}$ holds the sex, religion, age and education terms of Smith et al. (2014), the activity δ (five groups) and the homophily γ (White, Black, Hispanic, Asian) are solved by damped Newton so that the *expected* degree sum and the expected within-group ties of each group equal the GSS 2004 targets for the network's own nodes: mean degree 30 (Boase et al. 2006) times the relative number of confidants, and the share of same-race confidants. This moment matching is the maximum likelihood estimate of the dyad-independent model (what `ergm.ego` estimates for `nodefactor` + `nodematch(race, diff = TRUE)`); it converges in 9–11 iterations to a relative gap below 10^{-6}. "Other" (18 GSS respondents; 1.1% of the GSS nodes, 2.4% of the ATP nodes) is not calibrated: its degree target (0.49 $[0.25, 0.79]$ of the mean) is too noisy and homophily inside a residual category is not meaningful, so it gets relative degree 1 and the common race

coefficient. All four calibrated groups reproduce their targets inside the 95% bootstrap intervals (`02c_network_calibrated_tests.R`; in ATP the ratios between groups are those of the GSS, while degree relative to the ATP mean shifts with the composition).

Level x2. Degree-preserving swaps $(a,b),(c,d) \to (a,d),(c,b)$ that keep the race mixing matrix and the numbers of same-sex and same-religion ties *exactly* and the total age and education differences within 1%, accepted (Metropolis) toward twice the triangles of *every* node of x1, $\sum_i |t_i - 2\,t_i^{x1}|$, and then 5×10^7 more proposals that keep that error small while the network mixes (`02_closure_rewire.cpp`, `rewire_closure_node()`; a few seconds per network). The per-node error ends at 1.0%. x1 and x2 have the same nodes, degrees and mixing; they differ only in closure:

	GSS 2004 nodes		ATP 2014 nodes	
	x1	x2	x1	x2
transitivity	0.075	0.149	0.070	0.140
local clustering, lowest degree tercile	0.080	0.149	0.091	0.172
local clustering, middle tercile	0.080	0.159	0.075	0.149
local clustering, highest tercile	0.072	0.144	0.064	0.128
degree assortativity	0.31	0.39	0.26	0.32
mean distance	2.49	2.54	2.46	2.52
same race (share of ties)	0.893		0.887	
same sex	0.564		0.555	
same religion	0.688		0.597	
age difference (years)	11.72	11.83	12.18	12.30

Means over 24 networks; degree 30.0 on average (sd 12.5 GSS, 10.7 ATP; maximum 67 and 65). The age and education constraints bind: both totals end about 1% higher. x2 is not a draw from a probability model. It is the main set because real personal networks are clustered (81% of the pairs of GSS 2004 confidants know each other) and a graph of independent ties has almost no closure; the factor 2 is a choice, and it costs the premium a little (below), so it does not inflate the result.

Why every node, not the total. The first version of the rewiring targeted the total number of triangles (now level x2g, `-closure-mode=global`). The cheapest triangles are among high-degree nodes, so x2g concentrated them there: local clustering 0.08 in the lowest degree quintile and 0.21 in the highest, degree assortativity 0.33 → 0.46. The held-out GSS 2004 ego networks reject that profile (§8.2): closure among confidants is flat to falling with the size of the ego network, while in x2g it rises by 16–28 points. Doubling every node's triangles keeps the clustering–degree profile of x1 and reproduces the observed one.

Exploratory alternatives (`playground/PR4-rcpp-engine/generacion_de_redes.pdf`). An ERGM with GWESP could not be fitted with the race calibration: MCMLE did not converge, stochastic approximation reached the GWESP target but destabilized the small groups (Hispanics +36–41% off their targets; Asians bistable around a clique of 27 nodes), and ×3 degenerated with every method tried. `ergm.ego` with the GSS 2004 alter–alter ties did not converge either. The constrained rewiring keeps everything calibrated exactly and is cheap; what it gives up is a probability model for x2.

8.2 Held-out validation

`scripts/09_network_validation.R` splits the 1,424 respondents of the GSS 2004 network module in two halves stratified by race, computes the targets from half A, builds 24 networks per level from half-B nodes, and compares them with the confidants of half B (bootstrap of respondents and networks, $B = 1000$). Table and text for the paper in `paper-NDFU/sections/network_validation.tex`; figure in `plots/09_network_validation/`. In short:

- **Homophily.** Out of sample, relative degree and same-race share fall inside the intervals for every group (all egos: 0.896 vs. 0.895 $[0.874, 0.914]$). Not targeted: same religion -3% and age difference $+6\%$ (inside); same sex -5% $[-9, -1]$ and education difference $+9\%$ $[+2, +16]$ (small, outside). Random sets of k contacts are 30% more diverse in race than real confidants: the generator spreads cross-race ties evenly over egos.
- **Closure.** 80.5% of half-B confidant pairs know each other, 42.8% are especially close. Confidants are not a random draw from 30 contacts (random k: 8% in x1, 16% in x2). Taking the k most embedded contacts (most common neighbours with the ego), x2 falls inside that band (63%) and x1 at its lower edge (41%); taking the k nearest in Gower distance, 38% and 25%. x2 fits closure better than x1.
- **Profile.** Closure among confidants is flat to falling with the number of confidants (5+ minus 2–3: -4.7 $[-11.7, +1.9]$ points); x1 and x2 agree under all three rules, x2g does not (+16 to +28; every interval excludes zero).
- **Not reproduced.** In the networks, closure follows homophily: the most embedded contacts are 70–80% less diverse in race and religion than real confidants.

Caveat: the sex, religion, age and education coefficients come from Smith et al. (2014), who used the GSS 1985 and 2004 confidants, so those features are not independent of half B; the test is that the coefficients, the race calibration and 30 contacts together give the confidant-level shares.

8.3 Frozen gate and the engine

The gate. The pass probability of a tie is still $w(d_{ij}) = \sigma((h - d_{ij})/0.02)$. Under the *annealed* gate of §7.1 every frontier tie drew a new coin at every step, so a tie with small w still passed now and then: with enough steps the gate only delayed adoption, and the result depended on the step limit (200) and the patience (25); with 2000/400 the premium fell 7–10%. The *frozen* gate draws one coin per tie and simulation, shared by both directions: a tie is open or closed for the whole run, so the gate decides who counts. After the draw the cascade is deterministic and stops at a fixed point (about 6.5 steps on average); no step limit or patience is needed. This is the percolation reading of a stochastic threshold, in which thresholds count sources, not exposures (Centola & Macy 2007, p. 707 and n. 20).

The engine (`05_unified_engine.{cpp,R}`, Rcpp). `run_exposure_grid()` runs the whole (MSP, IUL) grid of one network, topology, λ and exposure condition in one call (0.13–0.15 ms per simulation on one core). `05_engine_check.R`: (1) the annealed engine reproduces the former R loop of 05, (2) the frozen engine an independent R cascade, (3) the sweep engine the frozen single-simulation engine, each in 2,376 of 2,376 cells; (4) the exposure formula equals hand values on a star ego; (5) every exposure condition equals an independent R implementation cell by cell. `05_netdiffuser_reference.R`: netdiffuseR 1.26.0 (`rdiffnet` on the open ties, with an adoption mechanism that divides by the full degree) gives the same number of adopters as the engine in 360 of 360 cells (about 200 ms against 0.15 ms per simulation).

The DP twin is rewired from the x2 network ($20m$ swaps, seed $500000 + k$) and is the *same* twin for x1 and x2, which have the same degrees: x2 − x1 is a paired comparison in which only the closure of the network changes. The seed cluster (0.40× maximum degree, primary shared by the network and its twin for random, central and marginal) and the simulation seeds are shared by x1 and x2 and by every exposure condition. In the CSV files the network is `topology = "NET"` (it was "GSS" until 2026-11, also for ATP).

8.4 The exposure axis (supplementary material)

The main text uses the base exposure $E_i = a_i/d_i$ (adopting contacts behind open ties over all contacts). The supplement varies how adopting contacts add up, with one formula and two dials ("C1" of `playground/PR3-similarity-vs-diversity/expresion_unica.pdf`):

$$E_i = \frac{\sum_g A_\theta(a_{ig}) + 2\kappa\, e_i^a}{\sum_g A_\theta(n_{ig}) + 2\kappa\, e_i^N}, \qquad A_\theta(x) = \frac{x}{1+\theta(x-1)},$$

where g runs over the circles of i's contacts (race × religion cells), a_{ig} and n_{ig} are the adopting (behind open ties) and all contacts of i in circle g, and e_i^a, e_i^N the ties among adopting contacts and among all contacts of i. Provisional names after Traag (2016): **independent reinforcement** θ (Kish's discount of redundant sources in a circle) and **cohesive reinforcement** κ (a bonus for every pair of adopting contacts who know each other). $\theta = \kappa = 0$ is the base. Both arms sit on one signed axis, $c =$ (value of the second adopter) -1: $c = -2\theta/(1+\theta)$ for θ, $c = 2\kappa$ for κ. The sweep uses $c = -1, -0.5$ ($\theta = 1, 1/3$), 0, $+0.5, +1$ ($\kappa = 0.25, 0.5$), and every arm level has a **permuted control** (same circle sizes, or the same number of pairs per ego, with random members): real − permuted is the part of the effect that comes from *where* the circles and pairs are. Results: `scripts/06c_exposure_axis.R`.

8.5 Results of the relaunch

`05_unified_diffusion_sweep_main.R` (2026-11-05): GSS and ATP, levels x2 and x1, five seeding strategies, nine exposure conditions, six λ, the 41×25 grid and 24 networks per set: $2 \times 2 \times 5 \times$ 2,656,800 = 53.1 million simulations in about 25 minutes on 8 cores; then 06, 06b ($B = 1000$), 06c, 06d and 07. All intervals are 95% run-bootstrap intervals (a run is a network, its twin and the primary seed draw; 24 per set). The tables below were generated from the CSV files.

The premium on the main networks (x2, base exposure, random seeding):

λ	β_{DP} (odds) te 12×12	odds drop	$\bar{\Phi}_{\mathrm{net}} - \bar{\Phi}_{\mathrm{DP}}$ (pp)	% of plane with gap > 0.10	MSP_c shift DP $-$ net
GSS 2004 nodes (GSS_x2)					
0.5	-0.644	-47%	5.02	22.7	0.100
	$[-0.665, -0.624]$		$[4.84, 5.18]$	$[22.0, 24.1]$	$[0.093, 0.103]$
0.6	-0.792	-55%	5.62	24.4	0.102
	$[-0.817, -0.772]$		$[5.46, 5.78]$	$[23.8, 25.2]$	$[0.095, 0.106]$
0.7	-0.975	-62%	6.39	26.1	0.101
	$[-1.007, -0.948]$		$[6.21, 6.56]$	$[24.6, 26.8]$	$[0.096, 0.108]$
0.8	-1.130	-68%	7.00	26.8	0.102
	$[-1.164, -1.102]$		$[6.78, 7.20]$	$[25.3, 27.3]$	$[0.095, 0.106]$
0.9	-1.256	-72%	7.50	27.4	0.099
	$[-1.284, -1.228]$		$[7.29, 7.71]$	$[26.1, 28.1]$	$[0.095, 0.105]$
1.0	-1.374	-75%	8.08	28.8	0.102
	$[-1.407, -1.342]$		$[7.84, 8.34]$	$[27.3, 29.3]$	$[0.095, 0.106]$
λ-mean	-1.028	-64%	6.60	26.0	0.101
	$[-1.054, -1.006]$		$[6.43, 6.76]$	$[25.2, 26.6]$	$[0.096, 0.104]$
ATP 2014 nodes (ATP_x2)					
0.5	-0.588	-44%	5.49	21.0	0.090
	$[-0.623, -0.551]$		$[5.08, 5.86]$	$[19.9, 22.1]$	$[0.075, 0.097]$
0.6	-0.706	-51%	6.01	21.9	0.090
	$[-0.745, -0.665]$		$[5.60, 6.39]$	$[20.9, 23.2]$	$[0.075, 0.100]$
0.7	-0.837	-57%	6.50	23.3	0.093
	$[-0.884, -0.788]$		$[6.07, 6.91]$	$[22.0, 24.5]$	$[0.080, 0.099]$
0.8	-0.991	-63%	7.10	24.5	0.094
	$[-1.042, -0.937]$		$[6.67, 7.52]$	$[23.4, 25.8]$	$[0.081, 0.102]$
0.9	-1.138	-68%	7.61	25.6	0.091
	$[-1.198, -1.080]$		$[7.17, 8.02]$	$[24.5, 26.7]$	$[0.082, 0.102]$
1.0	-1.287	-72%	8.06	26.3	0.097
	$[-1.354, -1.222]$		$[7.61, 8.49]$	$[24.7, 27.2]$	$[0.086, 0.109]$
λ-mean	-0.924	-60%	6.80	23.8	0.093
	$[-0.973, -0.874]$		$[6.37, 7.19]$	$[22.7, 24.8]$	$[0.080, 0.101]$

Reading. The premium survives the new networks and the frozen gate in both samples, at every λ, and no interval comes near zero. Rewiring the ties of x2 lowers the odds of adoption by 44–75% (GSS: λ-mean $\beta_{\mathrm{DP}} = -1.028$ $[-1.054, -1.006]$, probability scale 6.60 pp $[6.43, 6.76]$; ATP: -0.924 $[-0.973, -0.874]$ and 6.80 pp $[6.37, 7.19]$). It grows with λ as before (GSS 5.0 $\rightarrow$ 8.1 pp, ATP 5.5 $\rightarrow$ 8.1 pp). The rational channel is flat in λ (2.48 pp in GSS, 2.64 in ATP) and the social channel carries the growth (2.54 $\rightarrow$ 5.61 and 2.82 $\rightarrow$ 5.42 pp). DP needs about 0.1 more openness to tip (MSP_c 0.14 vs. 0.24 in GSS, 0.20 vs. 0.29 in ATP; ratios 1.73 and 1.47). The premium still needs the gate: with the gate nearly open ($h \geq 0.75$) the gap is -0.5 to 2.7 pp in GSS and 0.2 to 1.1 pp in ATP, so 91–100% of the plane mean comes from the part where the gate selects.

Against the previous design (96 ERGM networks, annealed gate, §7.1; GSS, random seeding, λ-mean). The premium is larger: 6.60 pp $[6.43, 6.76]$ against 5.46 $[5.30, 5.57]$, β_{DP} -1.028 against -0.851; the band covers 26.0% of the plane against 20.4%, and the tipping shift is 0.101 against 0.070. The exploratory runs (`playground/PR4-rcpp-engine`, $\lambda = 0.7$) attribute the increase to the frozen gate (about +1 pp: a closed tie never opens, so both topologies need more openness and the gap widens where the gate selects) and to the race calibration (about +0.5 pp), partly undone by closure (about -0.5 pp, below). The intervals for random and marginal seeding are five times narrower than before (width of β_{DP} 0.05 against 0.27), probably because small groups are no longer isolated, so where the primary seed lands matters less. The shape is unchanged: an L-shaped band along the tipping frontier, a flat rational channel, a stable tipping shift.

Seeding strategies (λ-mean):

	GSS_x2		ATP_x2	
strategy	β_{DP}	gap (pp)	β_{DP}	gap (pp)
central	−1.022	6.66	−0.942	6.97
	[−1.054, −0.988]	[6.45, 6.86]	[−0.969, −0.918]	[6.79, 7.16]
closeness	−1.060	6.87	−0.970	7.23
	[−1.095, −1.026]	[6.69, 7.07]	[−1.000, −0.943]	[7.04, 7.43]
eigen	−0.979	6.42	−0.923	6.85
	[−1.008, −0.950]	[6.25, 6.58]	[−0.957, −0.893]	[6.63, 7.07]
random	−1.028	6.60	−0.924	6.80
	[−1.054, −1.006]	[6.43, 6.76]	[−0.973, −0.874]	[6.37, 7.19]
marginal	−1.048	6.57	−0.973	7.24
	[−1.099, −0.980]	[6.11, 6.91]	[−1.005, −0.942]	[7.02, 7.45]

Reading. With the calibrated networks the hub strategies no longer beat random seeding: central − random is 0.06 pp $[-0.19, 0.33]$ in GSS and 0.18 $[-0.22, 0.60]$ in ATP (0.15 $[0.05, 0.31]$ with the ERGM networks). Closeness has the largest premium in both samples (closeness − random: 0.27 $[0.07, 0.52]$ and 0.44 $[0.00, 0.90]$), eigen the smallest in GSS (eigen − random −0.18 $[-0.38, 0.04]$), and random and marginal coincide. Every difference between strategies is below 0.5 pp, against a premium of 6.4–7.2 pp: the seeding strategy is a second-order factor.

What closure does: x2 vs. x1 (`06d_closure_contrast.R`; the same DP twin, so the difference is paired; λ-mean, pp; last column: the same difference over the selective half of the plane, $h \leq 0.5$):

	strategy	premium x2	premium x1	x2 − x1	relative	x2 − x1, $h \leq 0.5$
GSS	central	6.66	7.44	−0.78	−10.5%	−1.39
		[6.46, 6.86]	[7.29, 7.60]	[−1.01, −0.56]		[−1.79, −0.98]
GSS	closeness	6.87	7.55	−0.67	−8.9%	−1.22
		[6.69, 7.06]	[7.36, 7.73]	[−0.85, −0.51]		[−1.56, −0.88]
GSS	eigen	6.42	7.23	−0.81	−11.2%	−1.62
		[6.25, 6.59]	[7.08, 7.37]	[−0.99, −0.65]		[−1.91, −1.34]
GSS	random	6.60	7.14	−0.53	−7.5%	−0.81
		[6.43, 6.78]	[6.88, 7.36]	[−0.73, −0.33]		[−1.14, −0.48]
GSS	marginal	6.57	7.19	−0.62	−8.6%	−1.04
		[6.13, 6.93]	[6.96, 7.43]	[−1.02, −0.28]		[−1.79, −0.43]
ATP	central	6.97	7.66	−0.68	−8.9%	−1.29
		[6.77, 7.15]	[7.53, 7.79]	[−0.87, −0.50]		[−1.59, −0.98]
ATP	closeness	7.23	7.73	−0.50	−6.5%	−0.89
		[7.04, 7.42]	[7.57, 7.88]	[−0.66, −0.35]		[−1.18, −0.61]
ATP	eigen	6.85	7.56	−0.71	−9.4%	−1.30
		[6.62, 7.06]	[7.43, 7.69]	[−0.89, −0.53]		[−1.61, −1.00]
ATP	random	6.80	7.15	−0.35	−4.9%	−0.57
		[6.38, 7.18]	[6.72, 7.52]	[−0.72, 0.03]		[−1.25, 0.14]
ATP	marginal	7.24	7.38	−0.14	−1.9%	−0.24
		[7.02, 7.45]	[7.07, 7.64]	[−0.34, 0.05]		[−0.58, 0.10]

Reading. Closure lowers the premium, in both samples and every strategy: by 7–11% of the x1 premium in GSS (−0.53 pp $[-0.73, -0.33]$ with random seeding, −0.67 to −0.81 with the hub strategies) and by 2–9% in ATP, where the intervals for random and marginal seeding touch zero. The effect is 1.5–2 times larger in the selective half of the plane, where the gate filters. Choosing x2 over x1 makes the main result smaller, not larger. This is the direction of the GWESP draws of the exploration (more clustering, smaller premium): the premium comes from the homophily that x1 and x2 share, not from closure.

The exposure axis (`06c_exposure_axis.R`; random seeding; premium in pp; the permuted column is the premium of the arm's permuted control at $\lambda = 0.7$, and the last one real − permuted

at $\lambda = 0.7$):

condition	$\lambda = 0.5$	$\lambda = 0.7$	$\lambda = 1.0$	permuted, 0.7	real − permuted, 0.7
GSS_x2					
$\theta = 1$ ($c = -1.0$)	7.01 [6.75, 7.25]	8.56 [8.27, 8.83]	8.33 [8.06, 8.58]	8.71	-0.16 [−0.36, 0.06]
$\theta = 1/3$ ($c = -0.5$)	7.05 [6.84, 7.24]	8.72 [8.48, 8.93]	11.54 [11.21, 11.88]	9.59	-0.87 [−1.07, −0.67]
base ($c = +0.0$)	5.02 [4.85, 5.19]	6.39 [6.21, 6.58]	8.08 [7.82, 8.35]	–	–
$\kappa = 0.25$ ($c = +0.5$)	3.00 [2.83, 3.16]	3.18 [3.01, 3.35]	3.85 [3.65, 4.05]	2.84	0.34 [0.22, 0.47]
$\kappa = 0.5$ ($c = +1.0$)	2.79 [2.62, 2.95]	2.91 [2.75, 3.08]	3.39 [3.19, 3.57]	2.51	0.41 [0.32, 0.50]
ATP_x2					
$\theta = 1$ ($c = -1.0$)	7.25 [6.85, 7.62]	8.56 [8.14, 8.94]	8.86 [8.46, 9.23]	10.31	-1.75 [−1.92, −1.58]
$\theta = 1/3$ ($c = -0.5$)	6.70 [6.28, 7.08]	8.17 [7.74, 8.56]	9.50 [9.08, 9.88]	9.55	-1.37 [−1.51, −1.24]
base ($c = +0.0$)	5.49 [5.10, 5.84]	6.50 [6.07, 6.90]	8.06 [7.62, 8.48]	–	–
$\kappa = 0.25$ ($c = +0.5$)	3.57 [3.19, 3.91]	3.74 [3.35, 4.09]	4.17 [3.75, 4.56]	3.25	0.49 [0.41, 0.57]
$\kappa = 0.5$ ($c = +1.0$)	3.35 [2.97, 3.70]	3.38 [2.99, 3.73]	3.74 [3.33, 4.14]	2.76	0.61 [0.51, 0.70]

Reading. The premium never changes sign along the axis: 1.9 to 12.2 pp in GSS and 2.6 to 11.7 pp in ATP over every seeding, λ and condition (0 of 270 negative in each). Its size does depend on how adopting contacts add up. Independent reinforcement ($c < 0$) mostly raises it (−0.9 to +4.0 pp against the base in GSS, +0.5 to +2.1 in ATP) and cohesive reinforcement ($c > 0$) lowers it to 2.4–4.6 pp. But the permuted controls move it as much or more: with θ, real − permuted is −2.0 to +0.5 pp in GSS and −2.0 to −1.0 pp in ATP; with κ, +0.3 to +1.1 pp (ranges over every seeding and λ). The change comes from how many circles or pairs a set of contacts has, not from where they are in the network. The main text keeps the base exposure; this table goes to the supplement.

The reverse-premium corner (robust DP wins: one-sided max-T per strategy × λ panel, familywise 5%, as in §7.1). **GSS**: robust at $\lambda = 0.5$–0.7 in all five strategies (39–44, 32–44 and 8–18 robust corner cells), and deeper than with the ERGM networks: DP ahead by up to 11–15 pp, against 3–7 pp before. It is gone from $\lambda = 0.8$ (before: detectable but negligible there). **ATP**: only small robust cells (≤ 0.3 pp) at $\lambda \leq 0.7$, although single cells reach 3–5 pp. The corner stays a weak-coupling feature and, with these networks, a GSS one. A plausible reason for the deeper corner is the frozen gate (not isolated here): a tie that is closed never opens, so where the gate is wide open and agents are averse, homophily keeps the resistant among the resistant (the mechanism of §7.1).

Why the corner is a GSS feature: consolidation (`scripts/06e_consolidation.R`; x2, base exposure, random seeding, the seeds of 05). In GSS, aversion is tied to the Blau attributes: age, sex, education, race and religion explain 19% of the variance of q, and q is assortative over ties (0.076). In ATP they explain 2.7% (assortativity 0.022; q also has only 7 values, 10% of nodes at $q = 1$). Two manipulations separate association from distribution: permuting q across nodes (same distribution, no association) and mapping q by rank onto the other survey's distribution (same association):

survey	q	robust corner cells (DP ahead > 2 pp, $z > 3.5$) $\lambda = 0.5$	0.6	0.7
GSS	real	33	26	10
GSS	permuted	0	0	0
GSS	ATP distribution, same ranks	18	35	27
ATP	real	0	0	0
ATP	GSS distribution, same ranks	11	0	0

Permuting q removes the GSS corner; changing its distribution does not. The corner is a consolidation effect: when aversion is socially patterned, homophily seats the averse among the averse, and where the gate is open and λ low nobody reaches them; the DP twin mixes them with adopters. In ATP aversion is barely patterned, so there is no trap. The premium itself survives permuting q (GSS 5.2, 6.2, 6.9 pp at $\lambda = 0.5, 0.7, 1.0$, against 5.0, 6.4, 8.1 with the real q; ATP 5.2, 6.2, 7.4 against 5.5, 6.5, 8.1): it comes from homophily through the gate, and consolidation only shifts it (down at low λ, up at high λ). The same three checks (association, corner 2×2, premium with permuted q) are to be run on every new survey (ELSOC, GSS 2014).

What breaking the network destroys (`07_structural_diagnosis.R`; means over 24 networks, GSS; ATP within a few percent). From x2 to x1 only closure changes: global and local clustering halve ($0.149 \to 0.075$ and $0.151 \to 0.077$), degree assortativity falls from 0.39 to 0.31, modularity from 0.40 to 0.37, and the homophily on the five axes stays (race, sex and religion identical; age and education within 5%). The DP twin removes the rest: attribute assortativity drops to about 0 on every axis (race $0.69 \to 0.00$), the mean Gower distance over ties rises by 57% ($0.226 \to 0.355$), clustering falls to 0.039 (what the degree sequence alone produces), degree assortativity to 0 and modularity to 0.15. x2 $\to$ x1 $\to$ DP thus separates closure from homophily: closure moves the premium by 2–11% (downward), and the premium itself goes with the homophily that only DP removes.

Sensitivity to the level of closure: x3 (three times the triangles of every node: option `-closure=3` of 02; base exposure, five strategies, the same DP twin; 06d adds it when its sweep exists). The premium keeps falling as closure grows but stays far from zero. GSS, random seeding, λ-mean: x1 7.14, x2 6.60, x3 5.79 pp $[5.45, 6.08]$ (x3 $-$ x1 $= -1.34$ $[-1.65, -1.08]$, -19%); ATP: 7.15, 6.80, 6.13 pp $[5.54, 6.65]$ (-14%). Over the five strategies x3 is 10–20% below x1. The validation does not pick between x2 and x3 (both fall inside the GSS 2004 band with the embeddedness rule; x3 near its upper edge), so the level of closure belongs to the limitations: a factor of 2 costs the premium 2–11%, a factor of 3 costs 10–20%.

What the intervals cover now. The nodes are resampled for every network, so the run bootstrap covers the draw of agents from the survey's complete cases, the ties, the seed and the dynamics. It does not cover the sampling error of the GSS 2004 targets, the uncertainty of the coefficients of Smith et al. (2014), or the choice of the closure level (x2 vs. x3, below). The link-free gap remains the statistic to lead with in the paper; β_{DP} depends on the surface basis (§7.1).

Still open.

- The validation shows two small homophily misfits (same sex -5%, education difference $+9\%$) and that the generator ties closure to homophily; neither is expected to change the sign of the premium, but both belong in the limitations.
- "Other" (1–2% of the nodes) is not calibrated.
- The level of closure is a choice: x2 sits inside the GSS 2004 band only if confidants are the most embedded contacts.

9 To verify before writing any of this into the paper

- ☒ Exact reinforcement functional form in Tur 2024: $u_t = u_0 \cdot (1/a_t)^\gamma$ (multiplicative, power-law). *Resolved against both PDFs (2026-08-31):* Tur 2018 (PRE), Eq. (1), uses the **same** form (there called MQR, $q_t^i = q_0^i(1/a_t^i)^\gamma$); $a_t^i \in \mathbb{N}$ is explicitly the number of **adopting neighbors** (a state count, not received recommendations), with the rule evaluated only once informed ($a \geq 1$). Both papers restrict $\gamma \in [0, 1]$ — justified in 2018 by non-increasing marginal social influence (concavity), with the reductio that large γ trivializes diffusion on their $k = 4$ lattices — and sweep γ as a free "intensity" parameter (2024, Fig. 5: $\gamma \in \{0, 0.5, 1\}$; heatmaps over $[0, 1]$). **No empirical calibration of γ exists in either paper**; see the transportability caveat in §5.3.
- ☐ Whether Tur 2024's homophily-rewiring ρ has a formal analogue we should cite when claiming nesting (their $\rho \approx$ our imputation; not identical constructs).
- ☒ Whether the additive composition has more than conventional support. *Resolved 2026-11-10* (§5.6): it is the best-response form of conformity and coordination games; learning does not give Tur's form; no study measures the cross-partial; the premium survives both complementary forms.
- ☐ The random-utility framing sentence for IUL (already drafted in `JOURNAL_FIT_ANALYSIS.md`, §4d) composes cleanly with the situational-utility framing: $U_i(t) = \Gamma + \lambda \tilde{E}_i$ is a private valuation plus a social term — this is Brock & Durlauf (2001)'s decomposition (§5.7), worth citing.